

coverage. It demands no human intervention since a potentially unlimited energy is harnessed from the waves.

As for the impact such a technology could have, it could be significant, especially given how climate change has made the weather patterns erratic. The robot gives us a chance to manage the oceans better and help humans afford earlier alerts against phenomena such as tsunamis. It also brings better communication by which ships and fisheries could be managed better.

The bionic suit

Ekso Bionics works out of an old Ford manufacturing unit in California but there's nothing old about the bionic suit that they create. Indeed, it's a wave of the future that helps paraplegics walk again.

The suit was released in February of 2012. Within an year, the company marked about 1 million steps taken with the aid of the suit by some 500 patients. The company hopes to help a million people walking with the suit in a decade. Even though the current cost is less than easy-on-the-pocket: it's \$1,10,000, it's hoped that the price would plummet as the demand goes up.

The technology for the suit takes a leaf or two out of the page of load-carrying exoskeleton units which were originally developed for the military. However, Ekso's technology uses significantly less power (just 5

watts as opposed to 5,000 watts for the military units). Software and also the sensors included in the hip and knee regions of the device accurately assess when it's safe for the suit to take the next step, based on the wearer's body weight distribution. The suit could be sized to a particular patient in just five minutes. It has an anti-stumbling software – something which ensures at the very least 200 steps for the walkers using it for the first time.

This robotic technology, if it can be made sufficiently affordable could make the clunky wheelchairs obsolete.



Walk to the future